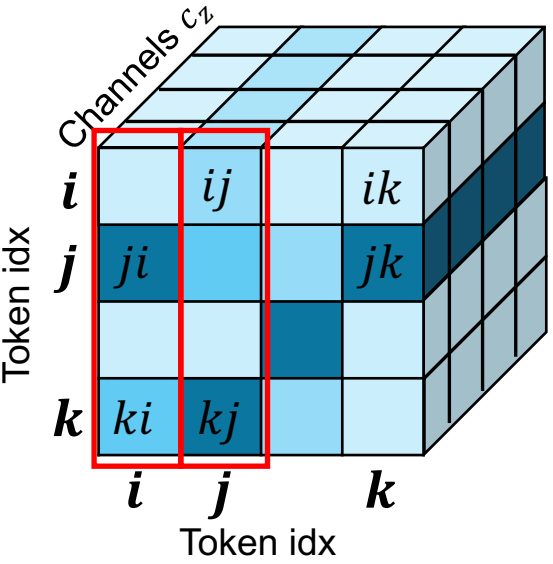
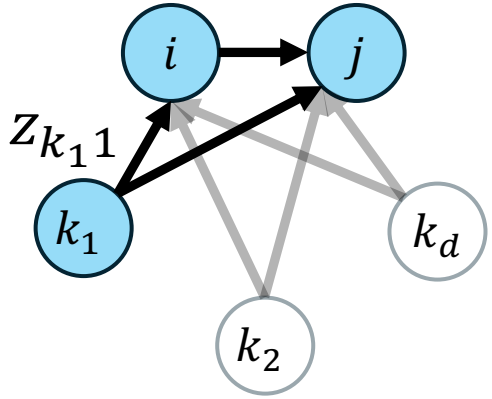
Pair representation z



Pair representations are directional

$$z_{ij} \neq z_{ji}$$

Updating z_{ij} using info. from incoming edges



$$d \in [1, N_{\text{tokens}}]$$

How does each k token relate to i and j tokens ?

$$d_{ij} < d_{k_1 i} + d_{k_1 j}$$

$$\vdots$$

$$d_{ij} < d_{k_d i} + d_{k_d j}$$

$$d_{ij} < \sum_k \left(d_{k_d i} + d_{k_d j}\right)$$

$$d_{ij} < \sum_k \left(d_{k_d i} \odot d_{k_d j}\right)$$